Session 19 · Thursday 29 October 2026 · Four to work. Everything on this sheet is about one thing: two genes with no wire between them still change each other.

Your circuit is not alone in there

Thursday's problem had a cure. A part has no boundary until you manufacture one — so manufacture it. Fix the junction, insulate, standardise, and the abstraction becomes true because you made it true.

Today's does not. Two genes with no regulatory connection change each other's output because they draw on the same ribosomes. You can fix a 5′UTR. **You cannot give your circuit its own ribosome pool.**

The one result the hour turns on. With $d_i = m_i/K_i$ the *demand* of gene i — how hard it pulls on the pool, message abundance and binding strength in one number —

$$p_i \;\propto\; \frac{d_i}{1+\sum_j d_j}$$

One weight on top, a sum over everything competing underneath. **You derived this shape on 15 September** for an operator that two factors compete for. The sum ran over configurations of one piece of DNA; here it runs over the genes of a cell.

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1 · The edge that is not in the diagram

Gene 1 runs at demand $d_1 = 1$. You induce gene 2.

(a) By what factor does gene 1's output fall when $d_2 = 4$? Do it in two lines.

(b) Write p_1 at demand d_2, divided by p_1 at $d_2 = 0$, and simplify until no p appears.

(c) That expression is a Hill function in d_2. What is its Hill coefficient, and what is its half-point K_{app}?

(d) So the fall is *exactly* the shape of repression by a non-cooperative repressor, and nothing is repressing anything. Given (c), design a measurement that distinguishes the two **without cloning anything**. One sentence, and say what you would vary.

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2 · The budget

(a) Normalise each gene's output by what it would reach **alone at unlimited demand**. Show that

$$p_1 + p_2 = \frac{\sum d}{1 + \sum d}$$

and say in one line why no d_1 or d_2 appears separately.

(b) You have a fixed total demand and you are free to split it between two genes however you like. What shape is the set of achievable (p_1, p_2)?

(c) The sum approaches 1 and never reaches it. Say what a designer is buying, and what they are paying, as they push the total demand up.

(d) Gyorgy *et al.* measured this and found something the one-pool model does **not** predict: induce a gene on a plasmid and a reporter on the *same plasmid* drops by up to 60%, while a reporter in the *chromosome* does not move. They also built a strain to show a chromosomal drop would have been visible. Why does that control matter, and what does the result say about where the pool ends?

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3 · The price, in growth

Scott *et al.* model a protein the cell has no use for as a neutral component and predict

$$\frac{\lambda}{\lambda_0} = 1 - \frac{\varphi_U}{\varphi_c}, \qquad \varphi_c \approx 0.48$$

where φ_U is the fraction of total protein given over to it.

(a) You express a useless protein at 10% of total protein. Estimate the growth-rate cost.

(b) Growth extrapolates to zero at $\varphi_U = 0.48$, not at 1.0. What is the other 52% doing, and why is that a ceiling rather than an accounting error?

(c) The model charges you for **proteome fraction and nothing else** — not amino acids, not ATP per peptide bond, not chaperone capacity. Their Fig. 4C puts three different proteins, from four expression systems, on the same relation. Why is that the evidence that licenses the simplification?

4 · The thing you already own, which was wrong

Session 5 gave you the response time of a protein:

$$t_{1/2} = \frac{\ln 2}{\gamma + \mu}$$

and PS2 used it with μ taken as a number the medium hands you.

(a) Your circuit switches on and diverts ribosomes. Which way does μ move, and which way does $t_{1/2}$ move?

(b) Which proteins does that affect?

(c) Nothing about the DNA changed. In two sentences: what does this do to the idea that a part has a response time you can look up?

(d) The direction here is unambiguous and the magnitude is not. Say what you would have to measure, in your own strain, before you could put a number on it.

The ledger — fill this in before the last ten minutes

You cannot insulate. So what *can* you do?

MOVE	WHAT IT COSTS YOU
lower the demand	
spend it better	
close a loop on burden itself	
characterise in place	
what you cannot do	

Before Tuesday

Fong & Palsson, *Nat Genet* 2004 — three pages in which a model predicts a growth rate, somebody builds the strain, evolves it, and checks. Read Supplementary Table 1 and the predicted-against-observed comparison: 39 of 50 right.

Come with a guess about what the eleven failures have in common.

PS7 goes out at the end of this session and is due 5 November. It covers sessions 17, 18 and 19.